Does the weight-aware base size carry to binary and Koblitz curves? (`ic-planner/3`)

2026-10-26 · `research/ic_planner_weight_rule_binkob_20261026` · protocol `1fffc5c45`, amendment 1 `2f87b17ad` · nine ecbench sessions, 1,568 records, all verified, audits OK.

On prime curves the cheapest base grows 1.41× at w = 1 GAE per unpriced native operation. Here: three binary and three Koblitz curves in Phase A, base multipliers ¼–4, with `G(m, w)` the geometric mean of `S_w(m)/S_w(plan)`. Phase B tests the Koblitz rule on three held-out curves.

Binary: no weight rule. `m* = 1` at every weight; at w = 1 the nearest alternative is `d₀ + 1` with G = 1.061.

Koblitz: the base shrinks. `m* = 0.71` at w ≥ 0.3 (G = 0.928 at w = 0.3, 0.847 at w = 1).

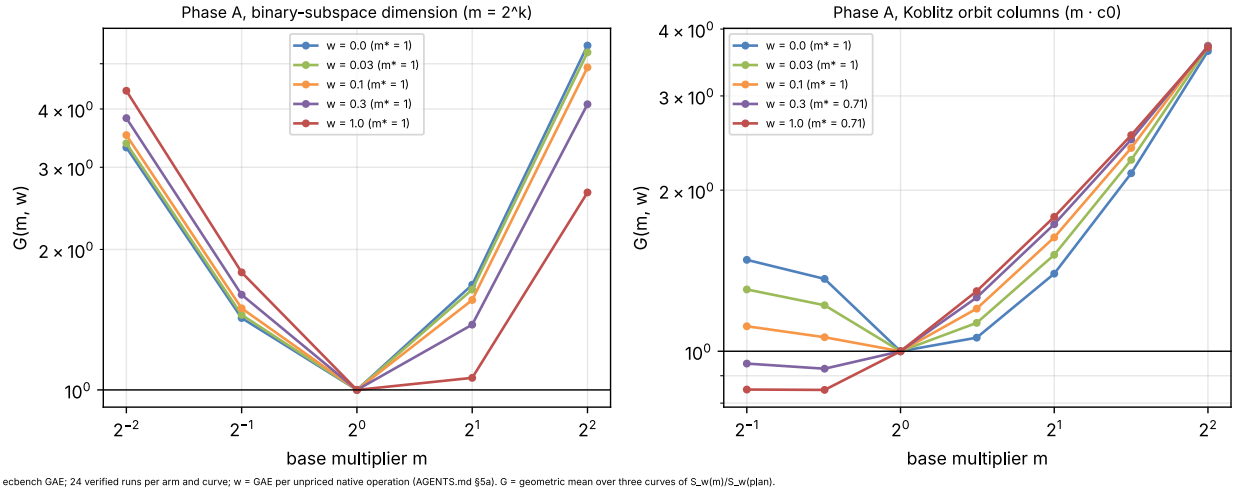


Figure 1: Phase A: G(m, w) per family.

held-out curve	plan ⟶ alt	w = 0	w = 0.3	w = 1
`icv1-f2m19-tm797-9c54981b`	2 ⟶ 1	1.587 [1.298, 1.939]	0.923 [0.790, 1.077]	0.802 [0.700, 0.919]
`icv1-f2m23-tm5197-1f85e9e1`	4 ⟶ 3	1.121 [1.022, 1.230]	0.928 [0.876, 0.983]	0.880 [0.840, 0.923]
`icv1-f2m39-t1481485-829ef1d1`	6 ⟶ 4	1.166 [1.043, 1.303]	0.895 [0.841, 0.952]	0.838 [0.797, 0.880]

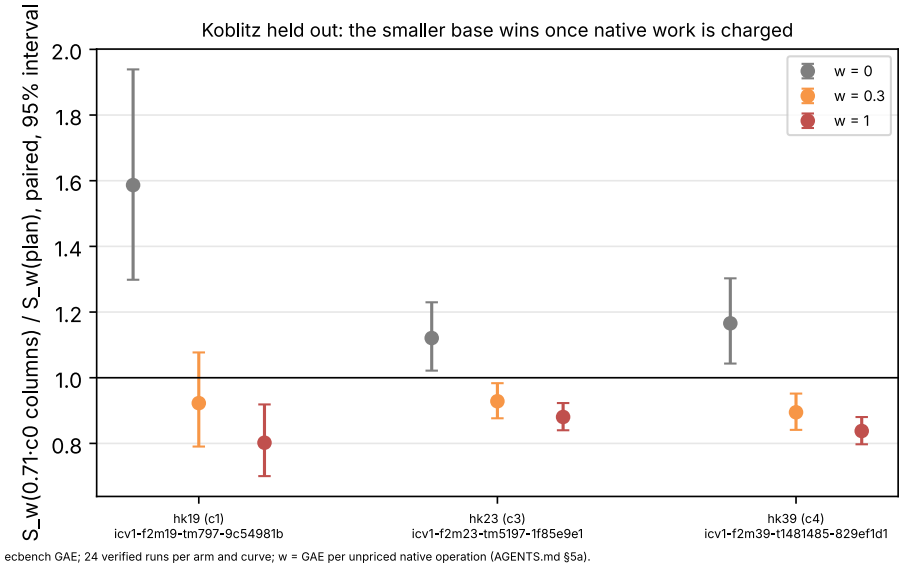


Figure 2: Koblitz held out, paired.

P3 held, with 2 of 3 intervals below 1 at w = 0.3 and 3 of 3 at w = 1. `ic-planner/3` gives `max(1, round(0.71·c₀))` Koblitz columns for w ≥ 0.3, and keeps the binary dimension with this study cited. The default (w = 0) plan is unchanged. P1 (non-decreasing `m*`) failed for Koblitz: a Koblitz trial's native work grows with the base, while a prime trial's does not.

The plan sits 2.6–4.8× rho in counted S on the held-out curves. Engineering class; no leaderboard row moves. Deviations: no resume checkpoint (no token; the owner said proceed), and the binary was built at `b561ebca8`, after the freeze (that change does not touch these arms' counts).